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Tiroshan Madushanka | Resume

I am a hard-working, passionate individual who has a deep interest towards new technological dimensions in the field of Computer Science.

- Status: MInf, BSc(Eng), Researcher, Lecturer, Software Engineer
- Fields: Data Science, Software Development, Education
- Stack: Python, Azure, Hadoop, Java, .Net
- Tools: Git, IntelliJ IDEA, Visual Studio, Terminal

Experience

Research Assistant (National Institute of Informatics (NII)- Japan) 2020 - Present

- Research on effect of Negative Sampling in Knowledge Graph Embedding
- Enhance Link Prediction by eliminating false negatives in negative sample space

Lecturer (University of Kelaniya - Sri Lanka) 2018 - Present

- Conduct lectures on Software Engineering concepts and theories
- Guide undergraduates on software engineering projects and research projects

Lead Technical Engineer (Rozie AI - USA /Sri Lanka) 2018 - Present

- Manage the micro service Infrastructure
- Lead DevOps and Back-end service Implementation

Senior Software Engineer (Zone 24x7 Inc. - USA /Sri Lanka) 2017- 2018

- Implemented Recommendation platform for KOHL's Department store, USA.
- Researched on transforming physical department store experience to online.

Software Engineer (Zone 24x7 Inc. - USA /Sri Lanka) 2015- 2017

- Implemented real-time IoT device monitoring and analytical system
- Researched on customer data analysis and data engineering.

Check out my projects at - **github.com/tiroshanm**

Education

Mater of Informatics (National Institute of Informatics, SOKENDAI) 2020 - 2022 Oct

- Master Thesis: Enhancing Link Prediction by Generating Quality Negative Samples
- Developed and evaluated an algorithm for enhancing Knowledge Graph Embedding

Bachelor of Science of Engineering (University of Moratuwa) 2010 - 2015

- Bachelor Thesis: Customer Churn Prediction by a Cognitive Approach
- Developed and evaluated an algorithm to enhance clustering

Certificate in JAVA Application Development (University of Colombo) 2010

- Implemented an application to manage university students
- Learnt concepts of JAVA Application Development

High School (Thurstan College, Colombo) 2006 - 2009

- Graduated high school in Physical Stream with rank 486
- Awarded scholarship for undergraduate studies in Faculty of Engineering, University of Moratuwa.

Projects

Effect of Negative Sampling in Knowledge Graph Embedding (NII)

- Introduced an algorithm to enhance the link prediction in knowledge graph embedding

Technology : python, pytorch, neo4j

Algorithm : Knowledge Graph Embedding

Natural Language Supporting for Customer Request Handling (Rozie AI)

- Implemented language models to understand natural conversations

Technology : python, Spacy models, scikit-learn, Azure, CosmosDB, Kubernetes, Docker

Algorithm : Logistic regression model, Conditional Random Fields Model.

The Customer Recommendation Generation Platform (Zone 24x7)

- Implemented core mathematical models & algorithms to generate customer recommendations

Technology : Hadoop, HBase, Hive, Map Reduce, Scala, Spark

Algorithm : Collaborative Filtering, Matrix Factorization, Word2vec

Customer Conversion Prediction System for Retail Industry (Zone 24x7)

- Implemented a model to generate accurate customer conversion predictions

Technology : Python, Pyspark, scikit-learn

Algorithm : Random Forest Classifier, Bagging, Gradient Boosting

Image Based Search System for Retail Industry (Zone 24x7)

- Implemented a platform for image based search adopting Neural Networks

Technology : Tensor Flow, Python

Algorithm : Convolutional Neural Networks

Analytical Platform for Customer Engagement (Zone 24x7)

- Implemented a platform to monitor customer interactions through PoS and IoT data)

Technology : Hadoop, HBase, Nifi, Kafka, Angular JS, Java

Algorithm : Clustering & Classification models

Publications

IEA/AIE Conference 2022

- **Madushanka,T.**, Ichise,R. MDNCaching: A strategy to generate quality negatives for knowledge graph embedding, Proceedings of International Conference on Industrial, Engineering & Other Applications of Applied Intelligent Systems, 2022.

AI-Cybersec@SGAI Conference 2021

- Ifthikar, A., Thennakoon, N., Malalgoda, S., Moraliyage, H., Jayawickrama, T., **Madushanka,T.**, & Hettiarachchi, S. A Novel Anomaly Detection Approach to Secure APIs from Cyberattacks, AI-Cybersec@SGAI, 2021.

ICACT Conference 2020

- Ranaweera, I., **Madushanka,T.**, Ranaweera, C. Building Automation System to Optimize Energy Utilization Acquiring the Best Performance of Appliances in a Distributed Network, Proceedings of International Conference on Advances in Computing and Technology, 2020

SCSE Conference 2020

- Harischandra, N.N., Jayakody, L.A., **Madushanka,T**. Impact of metacognition and age group on contemporary video game interface and gameplay design. Proceedings of International Research Conference on Smart Computing and Systems Engineering, 2020.

International Journal of Computer and Information Engineering 2014

- Senanayake, D., Muthugama, L., Mendis, L., **Madushanka,T**. Customer Churn Prediction: A Cognitive Approach. International World Academy of Science, Engineering and Technology, International Journal of Computer, Electrical, Automation, Control and Information Engineering, 9, 767-773, 2014.